



Migrating from STM32H5 to STM32C5 MCUs

Introduction

The STM32 microcontroller family, developed by STMicroelectronics, is known for its versatility and performance across a wide range of applications. As technology evolves, there is often a need to migrate from one microcontroller to another to meet new requirements or to optimize the cost and efficiency of the application.

This migration guide is designed to assist developers in transitioning from the STM32H5 series to the STM32C5 series. The STM32C5 series offers a cost-effective alternative to the STM32H5, making it an attractive option for applications where budget constraints are a significant consideration. Despite its lower cost, the STM32C5 series maintains a robust set of features and capabilities, ensuring that performance and functionality are not compromised.

1 Overview

Why migrate to STM32C5

There are several reasons why migrating from the STM32H5 to the STM32C5 series can be beneficial:

- **Cost reduction:** The STM32C5 series is designed to be a more affordable solution compared to the STM32H5 series. This cost reduction can be particularly advantageous in high-volume applications where every cent counts.
- **Adequate performance:** While the STM32C5 series is positioned as a lower-cost alternative, it still offers substantial computational power and peripheral integration. This ensures that applications can continue to perform efficiently without the need for excessive compromises.
- **Feature set:** The STM32C5 series includes a comprehensive set of peripherals and features that cater to a wide range of applications, from consumer electronics to industrial control systems.
- **Ease of migration:** STMicroelectronics provides extensive documentation and support to facilitate the migration process. This guide is part of that effort, offering detailed steps and considerations to ensure a smooth transition.

Scope of the guide

This guide provides a detailed comparison between the STM32H5 and STM32C5 series, highlighting the differences and similarities in their features and architecture. It covers the following key areas:

- **Hardware migration:** Details on package availability, pinout compatibility, and hardware considerations.
- **Boot mode compatibility:** Information on boot mode selection and system bootloader differences.
- **Peripheral migration:** A comprehensive comparison of the peripherals available in both series, including any changes in their architecture or functionality.
- **Software considerations:** Guidance on adapting software to accommodate the differences between the two series.

Target audience

This guide is intended for hardware and software engineers who are familiar with the STM32 microcontroller family and are considering or planning to migrate their applications from the STM32H5 to the STM32C5 series. A basic understanding of microcontroller architecture and embedded system design is assumed.

Additional resources

For further information, developers are encouraged to refer to the STM32H5 and STM32C5 datasheets and reference manuals. These documents provide in-depth technical details and are essential for a successful migration.

2 General information

STM32H5 and STM32C5 MCUs are 32-bit microcontrollers based on the Arm® Cortex® processor.

Note:

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Table 1. Reference documents

Document number	Title
[1]	STM32C5xx advanced Arm®-based 32-bit MCUs reference manual (RM0522)
[2]	STM32H523/33, STM32H562/63, and STM32H573 Arm®-based 32-bit MCUs reference manual (RM0481)
[3]	Arm® Cortex®-M33 32-bit MCU+FPU, 144 MHz, 512-Kbyte flash memory, 128-Kbyte RAM, math accelerators, datasheet (DSxxxxx)
[4]	Arm® Cortex®-M33 32-bit MCU+TrustZone®+FPU, 375 DMIPS 250 MHz, 512-Kbyte flash memory, 272-Kbyte RAM, math accelerators datasheet (DS14539 and DS14540)

3 STM32H5 and STM32C5 MCUs overview

3.1 Main features

The STM32C5 MCUs are designed to provide a cost-effective solution while maintaining the core essence and performance of the STM32 family. Although the STM32C5 series features a slightly reduced peripheral set compared to the STM32H5 series, it retains the essential capabilities and performance needed for a wide range of applications.

Key features of STM32C5 MCUs:

- Security:
 - Passive tamper and hide protection to support OEMiROT.
 - Simplified life cycle scheme with secure debug authentication
 - Additional encryption accelerator engines, including:
 - Advanced encryption hardware accelerator (AES)
 - Public key accelerator (PKA)
 - Secure AES coprocessor (SAES)
- Performance:
 - Frequency up to 144 MHz
 - Direct access to flash interface through ICACHE for internal and external memories
- Communication interfaces:
 - I3C, FDCAN, LPUART, USB, Ethernet
- Analog
 - ADC, COMP, OPAMP, DAC

This peripheral list ensures that developers can achieve adequate performance without compromising on the core functionality of their designs. The STM32C5 series provides a balanced mix of performance, security, and communication interfaces, making it suitable for a wide range of applications. Developers can continue to leverage the strengths of the STM32 family while benefiting from a more cost-effective solution.

The table below summarizes the memory availability of the STM32H5 and STM32C5 MCUs.

Table 2. Memory availability

Products	Flash memory		RAM size (Kbytes)				Feature level
	Size	Dual bank	SRAM1	SRAM2	SRAM3	BKPSRAM	
STM32H50x	Up to 128 Kbytes	Yes	16	16	NA	2	Without hardware crypto
STM32H533	Up to 512 Kbytes	Yes	128	80	64	2	With hardware crypto: AES, PKA,SAES, and OTFDEC
STM32H523	Up to 512 Kbytes	Yes	128	80	64	2	Without hardware crypto
STM32H573	Up to 2 MB	Yes	256	64	320	4	With hardware crypto: AES, PKA,SAES, and OTFDEC
STM32H563	Up to 2 MB	Yes	256	64	320	4	Without hardware crypto
STM32C54x	Up to 256 KB	Yes	32	32	NA	NA	With hardware crypto: AES
STM32C53x	Up to 256 KB	Yes	32	32	NA	NA	Without hardware crypto
STM32C56x	Up to 512 KB	Yes	64	64	NA	NA	With hardware crypto: AES
STM32C55x	Up to 512 KB	Yes	64	64	NA	NA	Without hardware crypto
STM32C5Ax	Up to 1 MB	Yes	128	128	NA	NA	With hardware crypto: AES, PKA and SAES
STM32C59x	Up to 1 MB	Yes	128	128	NA	NA	Without hardware crypto

3.2 System architecture

STM32C5 architecture is largely similar to STM32H5, with a few key differences:

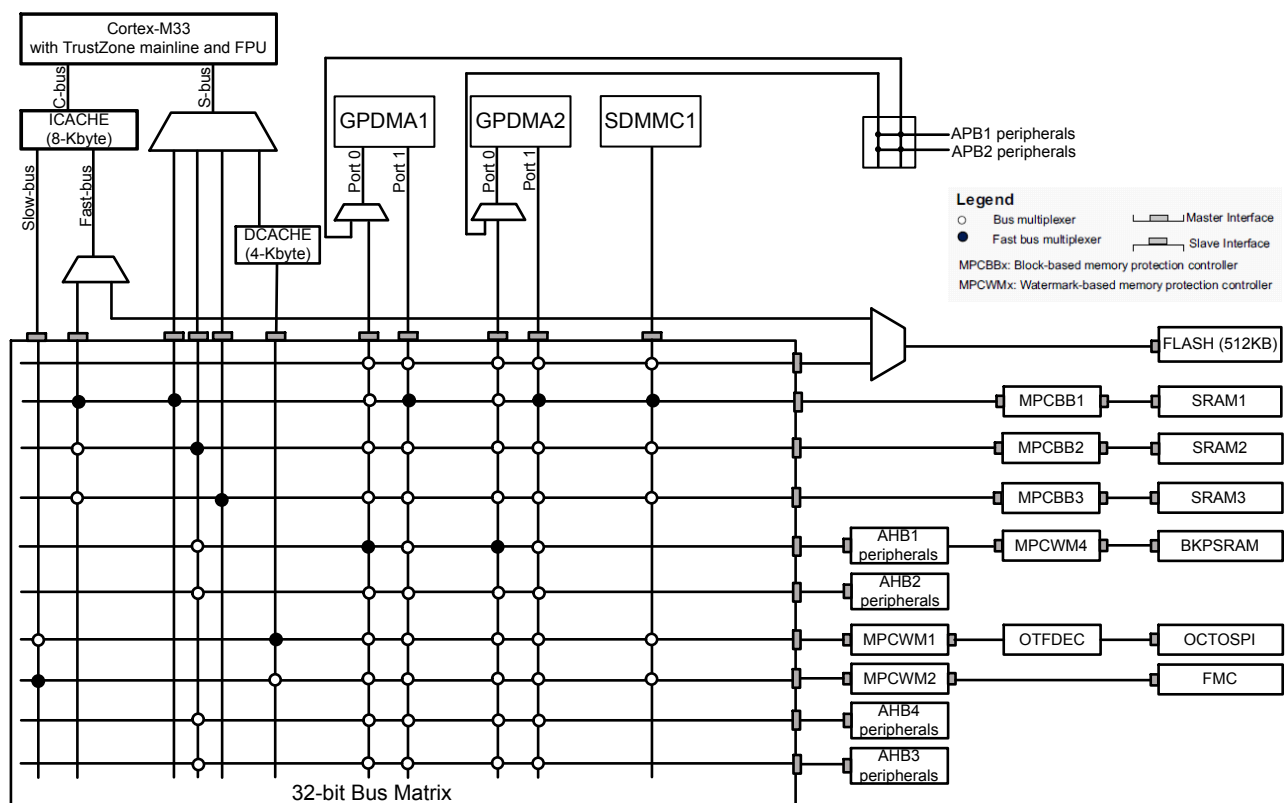
- High-speed memories: up to 1 MB dual bank flash memory and 256 Kbytes of SRAM.
- One Octo-SPI memory interface (one Octo-SPI available on all packages) and an extensive range of enhanced I/Os and peripherals connected to three APB buses.
- Three AHB buses and a 32-bit multi-AHB bus matrix.

A key difference between both implementation is replacement of the advanced GPDMA in STM32H5 by the more size optimised LPDMA.

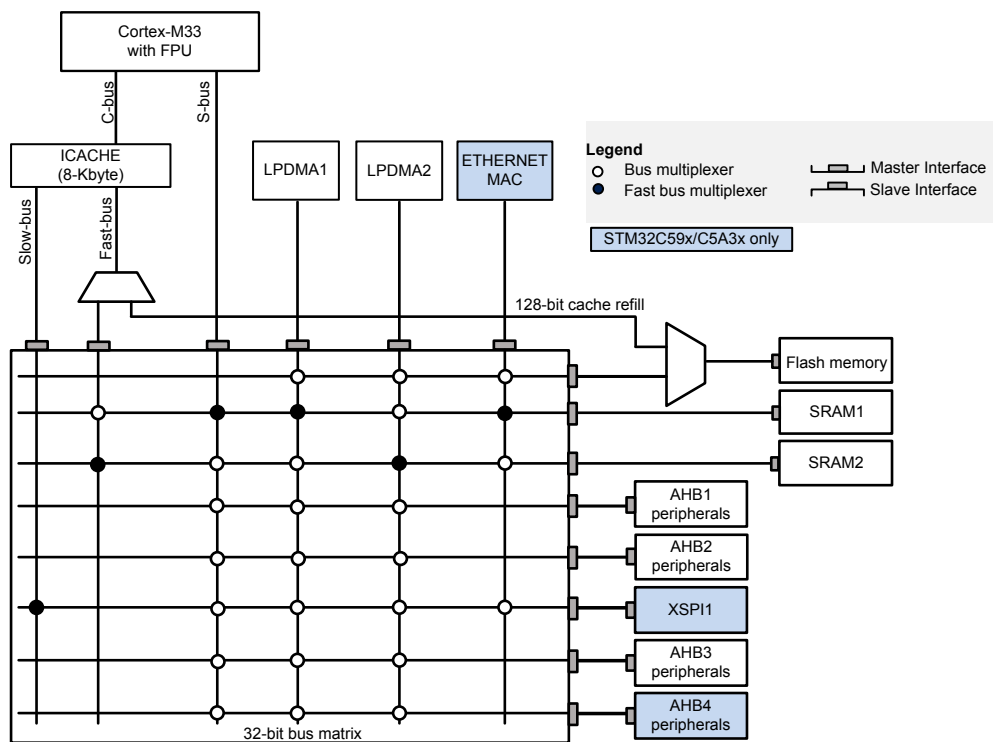
The bus matrix provides access from a controller to a target, enabling concurrent access and efficient operations even when several high-speed peripherals work simultaneously.

The figures below show the system architectures of STM32H523/533 and STM32C5.

Figure 1. STM32H533 system architecture



DT73447V2

Figure 2. STM32C5 system architecture


DTT6095V1

4 Hardware migration

This section presents the package and pinout compatibility details for the hardware migration.

4.1 Package availability

STM32H5 devices offer multiple packages from 20 to 225 pins.

The table below lists the available packages on STM32H5 devices compared to STM32C5 devices.

Table 3. Available packages

Package(size in mm x mm)	STM32H5	STM32C5
LQFP176 (24 x 24 mm)	Yes	No
LQFP144 (20 x 20 mm)	Yes	Yes
LQFP100 (14 x 14 mm)	Yes	Yes
LQFP80 (12 x 12 mm)	No	Yes
LQFP64 (10 x 10 mm)	Yes	Yes
LQFP48 (7 x 7 mm)	Yes	Yes
LQFP32 (7 x 7 mm)	No	Yes
UFQFPN48 (7 x 7 mm)	Yes	Yes
UFQFPN32 (5 x 5 mm)	Yes	Yes
UFQFPN24 (4 x 4 mm)	No	Yes
UFQFPN20 (3 x 3 mm)	No	Yes
TFBGA225 (13 x 13 mm)	Yes	No
UFBGA176+25 (10 x 10 mm, 0.80 mm pitch)	Yes	No
UFBGA169 (7 x 7 mm, 0.50 mm pitch)	Yes	No
UFBGA144 (10 x 10 mm, 0.80 mm pitch)	Yes	No
UFBGA144 (7 x 7 mm, 0.50 mm pitch)	Yes	No
UFBGA100 (7 x 7 mm, 0.50 mm pitch)	Yes	No
VFQFPN68 (8 x 8 mm)	Yes	No
TSSOP20 (6.5 x 4.4 mm)	No	Yes

1. X = available, NA = not available

4.2

Pinout compatibility

STM32C5 devices are not identical to STM32H5 devices in terms of MCU port assignment to package terminals, that is, in terms of pinout. This holds for all common package types of the packages listed in [Table 3. Available packages](#).

The following sections show the packages pinout figures and difference tables.

Table 4. Pinout compatibility

Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
1	VDD	VDD	VBAT	PE2	VBAT	PE2	VBAT	PE2	PE2	PE2	PE2	PE2
2	PC14-OSC32_IN(OSC32_IN)	PH0-OSC_IN(PH0)	PC13	PC13	PC13	PC13	PC13	PC13	PE3	PE3	PE3	PE3
3	PC15-OSC32_OUT(OSC32_OUT)	PH1-OSC_OUT(PH1)	PC14-OSC32_IN(OSC32_IN)	PC14-OSC32_IN(OSC32_IN)	PC14-OSC32_IN(OSC32_IN)	PC14-OSC32_IN(OSC32_IN)	PC14-OSC32_IN(OSC32_IN)	PC14-OSC32_IN(OSC32_IN)	PE4	PE4	PE4	PE4
4	NRST	NRST	PC15-OSC32_OUT(OSC32_OUT)	PC15-OSC32_OUT(OSC32_OUT)	PC15-OSC32_OUT(OSC32_OUT)	PC15-OSC32_OUT(OSC32_OUT)	PC15-OSC32_OUT(OSC32_OUT)	PC15-OSC32_OUT(OSC32_OUT)	PE5	PE5	PE5	PE5
5	VDDA	VREF+	PH0-OSC_IN(PH0)	PH0-OSC_IN(PH0)	PH0-OSC_IN(PH0)	PH0-OSC_IN(PH0)	PH0-OSC_IN(PH0)	PH0-OSC_IN(PH0)	PE6	PE6	PE6	PE6
6	PA0	PA0	PH1-OSC_OUT(PH1)	PH1-OSC_OUT(PH1)	PH1-OSC_OUT(PH1)	PH1-OSC_OUT(PH1)	PH1-OSC_OUT(PH1)	PH1-OSC_OUT(PH1)	VBAT	PH15	VBAT	PH15
7	PA1	PA1	NRST	NRST	NRST	NRST	NRST	NRST	PC13	PC13	PC13	PC13
8	PA2	PA2	VSSA	VREF-	VSSA	VREF-	PC0	PC0	PC14-OSC32_IN(OSC32_IN)	PC14-OSC32_IN(OSC32_IN)	PC14-OSC32_IN(OSC32_IN)	PC14-OSC32_IN(OSC32_IN)



Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
9	PA3	PA3	VDDA	VREF+	VDDA	VREF+	PC1	PC1	PC15-OSC32_OUT(OSC32_OUT)	PC15-OSC32_OUT(OSC32_OUT)	PC15-OSC32_OUT(OSC32_OUT)	PC15-OSC32_OUT(OSC32_OUT)
10	PA4	PA4	PA0	PA0	PA0	PA0	PC2	PC2	VSS	VSS	PF0	PF0
11	PA5	PA5	PA1	PA1	PA1	PA1	PC3	PC3	VDD	VDD	PF1	PF1
12	PA6	PA6	PA2	PA2	PA2	PA2	VSSA	VREF-	PH0-OSC_IN(PH0)	PH0-OSC_IN(PH0)	PF2	PF2
13	PA7	PA7	PA3	PA3	PA3	PA3	VDDA	VREF+	PH1-OSC_OUT(PH1)	PH1-OSC_OUT(PH1)	PF3	PF3
14	PB0	PB0	PA4	PA4	PA4	PA4	PA0	PA0	NRST	NRST	PF4	PF4
15	PB1	PB1	PA5	PA5	PA5	PA5	PA1	PA1	PC0	PC0	PF5	PF5
16	VCAP	VCAP	PA6	PA6	PA6	PA6	PA2	PA2	PC1	PC1	VSS	VSS
17	VDD	VDD	PA7	PA7	PA7	PA7	PA3	PA3	PC2	PC2	VDD	VDD
18	PB15	PB15	PB0	PB0	PB0	PB0	VSS	VSS	PC3	PC3	PF6	PF6
19	PA8	PA8	PB1	PB1	PB1	PB1	VDD	VDD	VSSA	PH4	PF7	PF7
20	PA9	PA9	PB2	PB2	PB2	PB2	PA4	PA4	VREF-	VREF-	PF8	PF8
21	PA11	PA11	PB10	PB10	PB10	PB10	PA5	PA5	VREF+	VREF+	PF9	PF9
22	PA12	PA12	VCAP	VCAP	VCAP	VCAP	PA6	PA6	VDDA	PH5	PF10	PF10
23	PA13(JTMS/SWDIO)	PA13(JTMS/SWDIO)	VSS	VSS	VSS	VSS	PA7	PA7	PA0	PA0	PH0-OSC_IN(PH0)	PH0-OSC_IN(PH0)
24	PA14(JTCK/SWCLK)	PA14(JTCK/SWCLK)	VDD	VDD	VDD	VDD	PC4	PC4	PA1	PA1	PH1-OSC_OUT(PH1)	PH1-OSC_OUT(PH1)



Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
25	PA15(JTDI)	PA15(JTDI)	PB12	PB12	PB12	PB12	PC5	PC5	PA2	PA2	NRST	NRST
26	PB3(JTDO/ TRACESWO)	PB3(JTDO/ TRACESWO)	PB13	PB13	PB13	PB13	PB0	PB0	PA3	PA3	PC0	PC0
27	PB4(NJTRS T)	PB4(NJTRS T)	PB14	PB14	PB14	PB14	PB1	PB1	VSS	VSS	PC1	PC1
28	PB5	PB5	PB15	PB15	PB15	PB15	PB2	PB2	VDD	VDD	PC2	PC2
29	PB6	PB6	PA8	PA8	PA8	PA8	PB10	PB10	PA4	PA4	PC3	PC3
30	PB7	PB7	PA9	PA9	PA9	PA9	VCAP	VCAP	PA5	PA5	VDD	PH4
31	BOOT0	PH2-BOOT0	PA10	PA10	PA10	PA10	VSS	VSS	PA6	PA6	VSSA	VREF-
32	PB8	PB8	PA11	PA11	PA11	PA11	VDD	VDD	PA7	PA7	VREF+	VREF+
33	-	-	PA12	PA12	PA12	PA12	PB12	PB12	PC4	PC4	VDDA	PH5
34	-	-	PA13(JTMS/ SWDIO)	PA13(JTMS/ SWDIO)	PA13(JTMS/ SWDIO)	PA13(JTMS/ SWDIO)	PB13	PB13	PC5	PC5	PA0	PA0
35	-	-	VSS	VSS	VSS	VSS	PB14	PB14	PB0	PB0	PA1	PA1
36	-	-	VDD	VDD	VDD	VDD	PB15	PB15	PB1	PB1	PA2	PA2
37	-	-	PA14(JTCK/ SWCLK)	PA14(JTCK/ SWCLK)	PA14(JTCK/ SWCLK)	PA14(JTCK/ SWCLK)	PC6	PC6	PB2	PB2	PA3	PA3
38	-	-	PA15(JTDI)	PA15(JTDI)	PA15(JTDI)	PA15(JTDI)	PC7	PC7	PE7	PE7	VSS	VSS
39	-	-	PB3(JTDO/ TRACESWO)	PB3(JTDO/ TRACESWO)	PB3(JTDO/ TRACESWO)	PB3(JTDO/ TRACESWO)	PC8	PC8	PE8	PE8	VDD	VDD
40	-	-	PB4(NJTRS T)	PB4(NJTRS T)	PB4(NJTRS T)	PB4(NJTRS T)	PC9	PC9	PE9	PE9	PA4	PA4
41	-	-	PB5	PB5	PB5	PB5	PA8	PA8	PE10	PE10	PA5	PA5



Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
42	-	-	PB6	PB6	PB6	PB6	PA9	PA9	PE11	PE11	PA6	PA6
43	-	-	PB7	PB7	PB7	PB7	PA10	PA10	PE12	PE12	PA7	PA7
44	-	-	BOOT0	PH2-BOOT0	BOOT0	PH2-BOOT0	PA11	PA11	PE13	PE13	PC4	PC4
45	-	-	PB8	PB8	PB8	PB8	PA12	PA12	PE14	PE14	PC5	PC5
46	-	-	VCAP	PB9	VCAP	PB9	PA13(JTMS/ SWDIO)	PA13(JTMS/ SWDIO)	PE15	PE15	PB0	PB0
47	-	-	VSS	VSS	VSS	VSS	VSS	VSS	PB10	PB10	PB1	PB1
48	-	-	VDD	VDD	VDD	VDD	VDD	VDD	VCAP	VCAP	PB2	PB2
49	-	-	-	-	-	-	PA14(JTCK/ SWCLK)	PA14(JTCK/ SWCLK)	VSS	VSS	PF11	PF11
50	-	-	-	-	-	-	PA15(JTDI)	PA15(JTDI)	VDD	VDD	PF12	PF12
51	-	-	-	-	-	-	PC10	PC10	PB12	PB12	VSS	VSS
52	-	-	-	-	-	-	PC11	PC11	PB13	PB13	VDD	VDD
53	-	-	-	-	-	-	PC12	PC12	PB14	PB14	PF13	PF13
54	-	-	-	-	-	-	PD2	PD2	PB15	PB15	PF14	PF14
55	-	-	-	-	-	-	PB3(JTDO/ TRACESWO)	PB3(JTDO/ TRACESWO)	PD8	PD8	PF15	PF15
56	-	-	-	-	-	-	PB4(NJTRS T)	PB4(NJTRS T)	PD9	PD9	PG0	PG0
57	-	-	-	-	-	-	PB5	PB5	PD10	PD10	PG1	PG1
58	-	-	-	-	-	-	PB6	PB6	PD11	PD11	PE7	PE7
59	-	-	-	-	-	-	PB7	PB7	PD12	PD12	PE8	PE8
60	-	-	-	-	-	-	BOOT0	PH2-BOOT0	PD13	PD13	PE9	PE9



Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
61	-	-	-	-	-	-	PB8	PB8	PD14	PD14	VSS	VSS
62	-	-	-	-	-	-	VCAP	PB9	PD15	PD15	VDD	VDD
63	-	-	-	-	-	-	VSS	VSS	PC6	PC6	PE10	PE10
64	-	-	-	-	-	-	VDD	VDD	PC7	PC7	PE11	PE11
65	-	-	-	-	-	-	-	-	PC8	PC8	PE12	PE12
66	-	-	-	-	-	-	-	-	PC9	PC9	PE13	PE13
67	-	-	-	-	-	-	-	-	PA8	PA8	PE14	PE14
68	-	-	-	-	-	-	-	-	PA9	PA9	PE15	PE15
69	-	-	-	-	-	-	-	-	PA10	PA10	PB10	PB10
70	-	-	-	-	-	-	-	-	PA11	PA11	VCAP	VCAP
71	-	-	-	-	-	-	-	-	PA12	PA12	VSS	VSS
72	-	-	-	-	-	-	-	-	PA13(JTMS/ SWDIO)	PA13(JTMS/ SWDIO)	VDD	VDD
73	-	-	-	-	-	-	-	-	VDDUSB	PH3	PB12	PB12
74	-	-	-	-	-	-	-	-	VSS	VSS	PB13	PB13
75	-	-	-	-	-	-	-	-	VDD	VDD	PB14	PB14
76	-	-	-	-	-	-	-	-	PA14(JTCK/ SWCLK)	PA14(JTCK/ SWCLK)	PB15	PB15
77	-	-	-	-	-	-	-	-	PA15 (JTDI)	PA15 (JTDI)	PD8	PD8
78	-	-	-	-	-	-	-	-	PC10	PC10	PD9	PD9
79	-	-	-	-	-	-	-	-	PC11	PC11	PD10	PD10
80	-	-	-	-	-	-	-	-	PC12	PC12	PD11	PD11



Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
81	-	-	-	-	-	-	-	-	PD0	PD0	PD12	PD12
82	-	-	-	-	-	-	-	-	PD1	PD1	PD13	PD13
83	-	-	-	-	-	-	-	-	PD2	PD2	VSS	VSS
84	-	-	-	-	-	-	-	-	PD3	PD3	VDD	VDD
85	-	-	-	-	-	-	-	-	PD4	PD4	PD14	PD14
86	-	-	-	-	-	-	-	-	PD5	PD5	PD15	PD15
87	-	-	-	-	-	-	-	-	PD6	PD6	PG2	PG2
88	-	-	-	-	-	-	-	-	PD7	PD7	PG3	PG3
89	-	-	-	-	-	-	-	-	PB3(JTDO/ TRACESWO)	PB3(JTDO/ TRACESWO)	PG4	PG4
90	-	-	-	-	-	-	-	-	PB4(NJTRS T)	PB4(NJTRS T)	PG5	PG5
91	-	-	-	-	-	-	-	-	PB5	PB5	PG6	PG6
92	-	-	-	-	-	-	-	-	PB6	PB6	PG7	PG7
93	-	-	-	-	-	-	-	-	PB7	PB7	PG8	PG8
94	-	-	-	-	-	-	-	-	BOOT0	PH2-BOOT0	VSS	VSS
95	-	-	-	-	-	-	-	-	PB8	PB8	VDD	VDD
96	-	-	-	-	-	-	-	-	PB9	PB9	PC6	PC6
97	-	-	-	-	-	-	-	-	PE0	PE0	PC7	PC7
98	-	-	-	-	-	-	-	-	VCAP	PE1	PC8	PC8
99	-	-	-	-	-	-	-	-	VSS	VSS	PC9	PC9
100	-	-	-	-	-	-	-	-	VDD	VDD	PA8	PA8



Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
101	-	-	-	-	-	-	-	-	-	-	PA9	PA9
102	-	-	-	-	-	-	-	-	-	-	PA10	PA10
103	-	-	-	-	-	-	-	-	-	-	PA11	PA11
104	-	-	-	-	-	-	-	-	-	-	PA12	PA12
105	-	-	-	-	-	-	-	-	-	-	PA13(JTMS/ SWDIO)	PA13(JTMS/ SWDIO)
106	-	-	-	-	-	-	-	-	-	-	VDDUSB	PH3
107	-	-	-	-	-	-	-	-	-	-	VSS	VSS
108	-	-	-	-	-	-	-	-	-	-	VDD	VDD
109	-	-	-	-	-	-	-	-	-	-	PA14(JTCK/ SWCLK)	PA14(JTCK/ SWCLK)
110	-	-	-	-	-	-	-	-	-	-	PA15 (JTDI)	PA15 (JTDI)
111	-	-	-	-	-	-	-	-	-	-	PC10	PC10
112	-	-	-	-	-	-	-	-	-	-	PC11	PC11
113	-	-	-	-	-	-	-	-	-	-	PC12	PC12
114	-	-	-	-	-	-	-	-	-	-	PD0	PD0
115	-	-	-	-	-	-	-	-	-	-	PD1	PD1
116	-	-	-	-	-	-	-	-	-	-	PD2	PD2
117	-	-	-	-	-	-	-	-	-	-	PD3	PD3
118	-	-	-	-	-	-	-	-	-	-	PD4	PD4
119	-	-	-	-	-	-	-	-	-	-	PD5	PD5
120	-	-	-	-	-	-	-	-	-	-	VSS	VSS



Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
121	-	-	-	-	-	-	-	-	-	-	VDDIO2	VDD
122	-	-	-	-	-	-	-	-	-	-	PD6	PD6
123	-	-	-	-	-	-	-	-	-	-	PD7	PD7
124	-	-	-	-	-	-	-	-	-	-	PG9	PG9
125	-	-	-	-	-	-	-	-	-	-	PG10	PG10
126	-	-	-	-	-	-	-	-	-	-	PG11	PG11
127	-	-	-	-	-	-	-	-	-	-	PG12	PG12
128	-	-	-	-	-	-	-	-	-	-	PG13	PG13
129	-	-	-	-	-	-	-	-	-	-	PG14	PG14
130	-	-	-	-	-	-	-	-	-	-	VSS	VSS
131	-	-	-	-	-	-	-	-	-	-	VDD	VDD
132	-	-	-	-	-	-	-	-	-	-	PG15	PG15
133	-	-	-	-	-	-	-	-	-	-	PB3(JTDO/ TRACESWO)	PB3(JTDO/ TRACESWO)
134	-	-	-	-	-	-	-	-	-	-	PB4(NJTRS T)	PB4(NJTRS T)
135	-	-	-	-	-	-	-	-	-	-	PB5	PB5
136	-	-	-	-	-	-	-	-	-	-	PB6	PB6
137	-	-	-	-	-	-	-	-	-	-	PB7	PB7
138	-	-	-	-	-	-	-	-	-	-	BOOT0	PH2-BOOT0
139	-	-	-	-	-	-	-	-	-	-	PB8	PB8
140	-	-	-	-	-	-	-	-	-	-	PB9	PB9



Pin	UFQFPN32		LQFP48		UFQFPN48		LQFP64		LQFP100		LQFP144	
	H503	C53x/C54x/C55x/C56x	H503/H523/H533	C53x/C54x/C55x/C56x/C59x/C5Ax	H503	C53x/C54x/C55x/C56x/C59x/C5Ax	H503/H523/H533/H56x/H573	C53x/C54x/C55x/C56x/C59x/C5Ax	H523/H533/H56x/H573	C55x/C56x/C59x/C5Ax	H523/H533	C59x/C5Ax
141	-	-	-	-	-	-	-	-	-	-	PE0	PE0
142	-	-	-	-	-	-	-	-	-	-	VCAP	PE1
143	-	-	-	-	-	-	-	-	-	-	VSS	VSS
144	-	-	-	-	-	-	-	-	-	-	VDD	VDD

5 Boot mode compatibility

5.1 Boot mode selection

STM32H5 and STM32C5 embed an SBS peripheral that controls boot and security features. For the STM32H5 device, the main boot control actions are listed below:

- Run the product with or without TrustZone® enabled.
- Choose between ST-iROT or OEM-iROT (refer to the reference manual for more details).
- Boot when launching a debug authentication sequence.
- Select boot between the bootloader or the user flash memory boot.
- Initialize the HDPL boot value.
In STM32C5 devices, the boot control action select boot between user flash or memory boot and initialize HDP boot value.

The boot configurations are chosen based on the product settings:

- **BOOT0**: to select booting on user flash memory or RSS (root secure services).
- **BOOT_UBE**: option byte to select the iROT between ST-iROT and OEM-iROT.
- **TZEN**: option byte to activate/deactivate the TrustZone®.
- **sbs_boot_addresses**: list of addresses defined by the flash memory:
 - NSBOOTADD: nonsecure boot address or BOOTADD on STM32C5 as no TrustZone®
 - SECBOOTADD: secure boot address
- **PRODUCT_STATE**: option byte to activate the different security mechanisms depending on the product use or **RDP_STATE** in STM32C5.
- **sbs_dbg_req**: used to launch the debug authentication protocol when booting.

The tables below present the STM32H533 boot modes, with TrustZone® either disabled or enabled.

Table 5. STM32H533 boot modes when TrustZone® is disabled (TZEN=0xC3)

PRODUCT_STATE	BOOT0 pin	BOOT_UBE FLASH_OPTSR[29:22]	Boot address option-byte selection	Boot area	STMicroelectronics default programmed value
Open	0	NA	NSBOOTADD[31:8]	Boot address defined by user option byte NSBOOTADD[31:8]	Flash: 0x0800 0000
	1	NA	NA	Bootloader	Bootloader
Provisioning	x	NA	NA	RSS	RSS
Provisioned, closed, locked	x	NA	NSBOOTADD[31:8]	Boot address defined by user option byte NSBOOTADD[31:8]	Flash: 0x0800 0000

Table 6. STM32H533 boot modes when TrustZone® is enabled (TZEN=0xB4)

PRODUCT_STATE	BOOT0 pin	BOOT_UBE FLASH_OPTSR[29:22]	Boot address option-byte selection	Boot area	STMicroelectronics default programmed value
Open	0	x	SECBOOTADD[31:8]	Boot address defined by user option byte SECBOOTADD[31:8]	Flash: 0x0C00 0000
	1	0xB4	NA	Bootloader	Bootloader
	1	0xC3	NA	ST-iROT	ST-iROT
Provisioning	x	NA	NA	RSS	RSS
Provisioned, TZ_Closed, closed, locked	x	0xC3	ST-iROT	ST-iROT	ST-iROT
	x	0xB4	SECBOOTADD[31:8]	Boot address defined by user option byte SECBOOTADD[31:8]	Flash: 0x0C00 0000

The tables below illustrate the STM32H523 boot modes, with TrustZone® either disabled or enabled.

Table 7. STM32H523 boot mode when TrustZone® is disabled (TZEN=0xC3)

PRODUCT_STATE	BOOT0 pin	Boot address option- byte selection	Boot area	STMicroelectronics default programmed value
Open	0	NSBOOTADD[31:8]	Boot address defined by user option byte NSBOOTADD[31:8]	Flash: 0x0800 0000
	1	NA	Bootloader	Bootloader
Provisioning	x	NA	RSS	RSS
Provisioned, closed, locked	x	NSBOOTADD[31:8]	Boot address defined by user option byte NSBOOTADD[31:8]	Flash: 0x0800 0000

Table 8. STM32H523 boot mode when TrustZone® is enabled (TZEN=0xB4)

PRODUCT_STATE	BOOT0 pin	Boot address option- byte selection	Boot area	STMicroelectronics default programmed value
Open	0	SECBOOTADD[31:8]	Boot address defined by user option byte SECBOOTADD[31:8]	Flash: 0x0C00 0000
	1	NA	Bootloader	Bootloader
Provisioning	x	NA	RSS	RSS
Provisioned, TZ_Closed, closed, locked	x	SECBOOTADD[31:8]	Boot address defined by user option byte SECBOOTADD[31:8]	Flash: 0x0C00 0000

The table below represent the boot modes for the STM32C5 series.

Table 9. STM32C5 boot modes

RDP_STATE	BOOT_SEL option bit	BOOT0 option bit	BOOT0 pin	EMPTY flag value ⁽¹⁾	Boot area
Level 0	1	x	0	x	Boot address defined by user option byte BOOTADD[31:8]

RDP_STATE	BOOT_SEL option bit	BOOT0 option bit	BOOT0 pin	EMPTY flag value ⁽¹⁾	Boot area
Level 0	1	x	1	x	Bootloader
	0	0	x	0	Boot address defined by user option byte BOOTADD[31:8]
				1	Bootloader
		1		x	Bootloader
Level 2 with boundary scan and level 2	x	x	x	x	Boot address defined by use option byte BOOTADD[31:8]

1. The boot address, defined by BOOTADD option byte, is considered empty only if the FLASH word is equal to 0xFFFF FFFF. In this case, at option byte loading, the EMPTY bit is set in FLASH_ACR.

5.1.1

Empty check

During the option byte loading phase, after loading all options, the flash memory interface checks whether the boot location in the main memory, indicated by BOOTADD[31:8], is programmed. The result of this check, with the boot0 information, is used to determine where the system should boot from. It prevents the system from booting from the main flash memory area when, for instance, no user code has been programmed.

The main flash memory empty check status can be read from the EMPTY bit in the FLASH access control register (FLASH_ACR). Software can modify it by writing an appropriate value to the EMPTY bit. The BOOT_LOCK, which locks modifications of option bytes involved in BOOT selection, has no impact on this empty check mechanism.

5.2 System bootloader

The system bootloader is located in the system memory, programmed by STMicroelectronics in production. It is used to reprogram the flash memory using one of the serial interfaces.

The following table shows the supported communication peripherals by the system bootloader. For more details, refer to the application note *STM32 microcontroller system memory boot mode* (AN2606).

Table 10. Bootloader communication peripherals

System bootloader peripherals	STM32H5 I/O pin	STM32C5 I/O pin
DFU ⁽¹⁾	PA11/PA12	
USART1	PA10/PA9	
USART2	PA2/PA3	
USART3	PD9/PD8 ⁽²⁾	
CAN	FDCAN2(PB5/PB13) ⁽³⁾	PB5/PB6 ⁽⁸⁾
I2C1	PB8/PB9 ⁽⁴⁾	NA
I2C2	NA	NA
I2C3	PA8/PC9 ⁽⁵⁾	NA
I3C1	PB6 ⁽⁶⁾ /PB7	NA
SPI1	PA7/PA6/PA5/PA4	PA7/PA6/PA5/PA4
SPI2	PC1 ⁽⁷⁾ /PB14/PB10/PB12	PB15/PB14/PB13/PB12
SPI3	PC12/PC11/PC10/PA15 ⁽⁴⁾	PB0/PB1/PB2/PB8 ⁽⁹⁾
SPI4	NA	PE14/PE13/PE12/PE11

1. On STM32H5, the USB DFU bootloader does not need an external quartz. It uses internal HSI48.

2. Only for LQFP100, LQFP144, UFBGA144 and UFBGA100.

3. On STM32H5xx, the FDCAN bootloader does not use an external quartz. It uses HSI and PLL.

4. Not available for LQFP48, UFQFN48, WLCSP39.

5. PC9 is replaced by PB4 on LQFP48, UFQFN48 and WLCSP39.

6. PB6 is replaced by PB8 on LQFP48, UFQFN48 and WLCSP39.

7. PC1 is replaced by PB15 on LQFP48, UFQFN48 and WLCSP39.

8. FDCAN1 on STM32C55x/C56x and FDCAN2 on STM32C53x/C4x/C59x/C5Ax.

9. Not supported on STM32C53x/C54x.

6 Peripheral migration

6.1 Cross compatibility between STM32 products

STM32 microcontrollers embed a set of peripherals that can be classified in the following groups:

- Group 1: peripherals common to all products.
These peripherals are identical, so they have the same structure, registers, and control bits. All functionalities can be retrieved after migration at application level. All features and behaviors remain the same.
- Group 2: peripherals shared by all products, but with minor differences (to support new features, in general).
The migration from one product to another is easy and does not require any significant new development efforts.
- Group 3: peripherals that have considerable changes from one product to another (new architecture or new features, for example).

For this group of peripherals, the migration requires a new development at the application level.

For STM32H523/533, all of the following can be configured as trusted or untrusted: each GPIO or peripheral, DMA channel, clock configuration register, ICACHE, DCACHE, and small part of the flash memory or SRAM. The following table summarizes the available peripherals in STM32C5 compared to STM32H523/533.

Table 11. STM32 peripheral compatibility between products

Peripherals		STM32H523/533 ⁽¹⁾	STM32C5 ⁽¹⁾
Core		Cortex®-M33 with TrustZone® mainline and FPU	Cortex®-M33 with FPU
ST immutable root of trust (STiROT)		Yes	No
Maximum CPU frequency		Up to 250 MHz	Up to 144 MHz
Flash memory		Up to 2 Mbytes, with high-data-cycle capability	Up to 1 Mbytes with extra 64 KB
SRAMs	System	640 Kbytes (256+64 (w/ECC)+320 (w/ flex ECC))	256 Kbytes (128 (w/ECC) + 128)
	Backup	Up to 4 Kbytes	NA
Timers	General purpose	2 (32 bits) and 8 (16 bits)	2 (32 bits) and 4 (16 bits)
	Advanced control	2 (16 bits)	2(16 bits)
	Basic	2 (16 bits)	2 (16 bits)
	Low power	6 (16 bits)	1 (16 bits)
	SysTick timer	2	1
	Watchdog timers (independent, window)	2	2
Communication interfaces	SPI/I2S	Up to 6 SPIs, including three muxed full-duplex I2S, and up to 6 additional SPIs from USART when configured in Synchronous mode (one additional SPI with Octo-SPI)	3 with muxed full-duplex I2S and up to 4 additional SPIs from USART when configured in synchronous mode
	I2C	4 (Sm, Fm, and Fm+ interfaces (SMBus/PMBus)	3 (sm, fm, and fm+ interfaces (SMBus/PMBus)
	I3C	2	1
	USART/UART	6/6	4/2
	LPUART	1	1
	USB	USB FS	USB FS
	UCPD	Yes	No

Peripherals		STM32H523/533 ⁽¹⁾	STM32C5 ⁽¹⁾
Communication interfaces	CAN	2 FDCAN	2
	SAI	2	No
	SDIO	2	No
	DCMI	Yes	No
	PSSI	Yes	No
	ETHERNET	Yes	Yes
Flexible memory controller (FMC)		Yes (8-, 16-bit data bus width); no SDRAM	No
Octo-SPI		1	1
Quad SPI memory interface (Quad-SPI)		Yes (Octo-SPI in quad-mode)	
HDMI-CEC		Yes	No
CRC		Yes	Yes
DMA		2 GPDMA (featuring two controller ports), TrustZone® support/linked-list	2 LPDMA (up to 8 streams each)
Real-time clock (RTC)		Yes	Yes
Random number generator (RNG)		Yes	Yes
SAES, AES		Yes ⁽²⁾	Yes ⁽²⁾
Public key accelerator (PKA)		Yes	Yes
HASH (SHA-512)		Yes	Yes
On-the-fly decryption engine (OTFDEC)		Yes ⁽²⁾	No
GPIOs		Up to 140	Up to 118
ADC (12 bits)		2 (12-bit ADC with up to 5 MSPS); up to 20 channels	3 (12-bit ADC 2.25 MSPS and 4.5 MSPS in dual interleaved mode); up to 28 channels
DAC (12 bits)		1 (two channels)	1 (Up to two channel)
Digital temperature sensor		Yes	No
Operating temperatures		Ambient operating temperature: -40 to +85°C/-40 to +125°C Junction temperature: -40 to +130°C	Ambient operating temperature: -40 to +85°C/-40 to +125°C Junction temperature: -40 to +140°C
Operating voltage		1.71 to 3.6 V	2.7 to 3.6 V
Internal voltage reference buffer		Yes	No

1. For more details, refer to the product datasheets.
2. Only for product supporting cryptographic peripherals.

6.2 Migration of system peripherals

6.2.1 Embedded flash memory (FLASH)

The following table compares the flash memory interface on STM32C5 and STM32H523/533 devices.

Table 12. FLASH features

Flash memory	STM32H5	STM32C5
Main/program memory	- Up to 1 Mbytes in STM32C5, up to 4 Mbytes in STM32H5 (dual bank) - Flash memory read operations supporting multiple lengths: 128 bits, 64 bits, 32 bits, 16 bits, or one byte 8 Kbytes sector erase, bank erase and dual-bank mass erase	
Dual bank	Read while write (RWW) and flash bank swapping	
Error code correction (ECC)	One error detection/correction or two error detections per 128-bit flash memory word using 9 ECC bits	
Wait states	Up to 6 (depending on the supply voltage and frequency)	
One time programmable (OTP) memory	2 Kbytes (OTP) area	4.5 Kbytes (OTP) for user data
FLASH security and protections	- TrustZone® backed watermark and block security protection - HDP protection providing temporal isolation - Configuration protection - Write protection - Device nonvolatile security life cycle and application boot state management	- Readout protection (RDP) - HDP protection providing temporal isolation - Configuration protection - Write protection - Device nonvolatile application boot state management
User option bytes ⁽¹⁾	nRST_STBY nRST_STOP IWDG_SW WWDG_SW IWDG_STBY, IWDG_STOP BOR_LEV BORH_EN BOOT_UBE OPTSTRT OPTLOCK WRPSG PRODUCT_STATE IO_VDDIO2_HSLV SWAP_BANK	nRST_STBY nRST_STOP IWDG_SW WWDG_SW IWDG_STBY, IWDG_STOP BOOT0 BOOTSEL SINGLE_BANK BOOT_UBE WRPSG RDP_LEVEL SWAP_BANK

1. Refer to the "Option-byte organization" table in the reference manual that provides all user option bytes.

6.2.2

SRAMs

The RAMCFG controller, a new peripheral available on STM32H5, is dedicated to control SRAM1, SRAM2, SRAM3, and BKPSRAM. Refer to the *RAMs configuration controller* section in the corresponding reference manual for more details.

Table 13. SRAM features

Features	STM32H5	STM32C5
Size	Up to 640 Kbytes: 256-Kbyte SRAM1 64-Kbyte SRAM2 w/ECC 320-Kbyte SRAM3w/ flex ECC 4-Kbyte BKPSRAM	Up to 256 Kbytes: 128-Kbyte SRAM1 128-Kbyte SRAM2 w/ECC
Access by DMA and CPU	Bytes, half-words (16 bits), or full words (32 bits) possible access.	Bytes, half-words (16 bits), or full words (32 bits) possible access.
CPU access bus	- System bus or C-bus. - BKPSRAM (only system bus).	System bus or C-bus.
Retention	BKPSRAM: - Optional retention in Standby mode. - Optional retention in VBAT mode.	SRAM1 and SRAM2 can be optionally retained under stop mode down to a 16KB block only.
Security	- When the TrustZone® security is enabled, all SRAMs are secure after reset. - The SRAMs can be programmed as nonsecure, using the MPCBB with a block granularity of 512 bytes.	NA
Hardware and software erase conditions	- SRAM1 and SRAM2 erase can be requested by executing a specific software sequence, detailed in section <i>RAMCFG</i> of the product reference manual. - SRAM2 and optional backup SRAM are protected by the tamper detection circuit, and erased by hardware in case of tamper detection. - SRAM2 is erased in case of regression.	
System reset erase	- SRAM2 can be erased with a system reset using the option bit SRAM2_RST option bit in the flash memory user option bytes. - SRAM1 and SRAM3 are erased when a system reset occurs if the SRAM13_RST option bit is selected in the flash memory user option bytes (SRAM1_RST in STM32C5 devices).	
Error detection and correction	- Single error detection and correction with interrupt generation. - Double error detection with interrupt or NMI generation. - The ECC is supported by SRAM2 and BKPSRAM when enabled with the SRAM2_ECC and BKPRAM_ECC user option bits. - ECC: 7 bits are added per 32 bits. - Interrupts are generated when single- and/or double-ECC errors are detected: - Two ECC RAMCFG interrupts. - One ECC NMI interrupt.	
Write protection	- SRAM2 can be write-protected with a page granularity of 1 Kbyte. - Each 1-Kbyte page can be write-protected by setting its corresponding PxWP (x = 0 to 63) bit in RAMCFG registers.	

6.2.3

Direct memory access controller (DMA)

STM32C5 and STM32H523/533 have different DMA architectures and features.

All devices embed two DMA controllers:

- LPDMA1 (eight channels) and LPDMA2 (eight channels) for STM32C5.
- GPDMA1 (eight channels) and GPDMA2 (eight channels) for STM32H523/533.
Each GPDMA and LPDMA instance has the same channel-based implementation and is connected to the same requests and triggers. In comparison to GPDMA, LPDMA do not implement FIFO, consequently does not support burst mode.

The following table illustrates the main differences between DMA requests in STM32C5 and STM32H523/533.

Table 14. DMA features Peripherals

Peripherals	STM32H5		STM32C5	
	GPDMA1	GPDMA2	LPDMA1	LPDMA2
Architecture	Each instance of DMA controllers can access memory and peripherals			
Number of instances	1	1	1	1
Number of controllers	Dual bidirectional AHB controller		Single AHB controller bus	
Number of channels	8	8	8	8
TrustZone® security	Yes		NA	
Per chanel FIFO				
Privileged/unprivileged DMA	Yes			
Linked-list				

6.2.4

Reset and clock control (RCC)

The table below presents the main differences related to the RCC (reset and clock controller) between STM32C5 and STM32H523/533 devices.

Table 15. RCC features

RCC	STM32H5	STM32C5
HSI	64 MHz RC oscillator	144 MHz oscillator 3 outputs clock possible: • HSI144 144 MHz intended for SYSCLK • HSI144DIV3 48 MHz intended for USB • HSI144 prescaled clock for peripheral kernel clock
PSI	NA	100 MHz, 144 MHz or 160 MHz oscillator 3 outputs clock possible: • PSIS100 100 MHz or 144 MHz intended for SYSCLK • PSIDIV3 intended for USB • PSIK prescaled clock for peripheral kernel clock
CSI	CSI: low-power RC oscillator that can be used directly as a system clock, peripheral clock, or PLL input: • Low-cost clock source since no external crystal is required • Faster startup time than HSI (a few microseconds) • Very low-power consumption • The CSI provides a clock frequency of about 4 MHz	NA
HSI48	48 MHz RC oscillator HSI48 can drive USB and RNG.	NA

RCC	STM32H5	STM32C5
LSI	32 kHz RC Lower consumption, higher accuracy	
HSE	From 4 to 50 MHz	From 4 to 50 MHz
LSE	32.768 kHz Configurable drive/consumption	32.768 kHz Configurable drive/consumption
PLL	Three PLLs: - Main PLL (PLL1) provides clocks for CPU and some peripherals - PLL2 and PLL3 generate the kernel clock for peripherals Each PLL offers three outputs with postdividers. Input frequency range: 2 to 16 MHz for the VCO in wide-range mode 1 to 2 MHz for the VCO in low-range mode	No PLLs
AHB frequency	Up to 250 MHz	Up to 144 MHz
APB1 frequency		
APB2 frequency		
RTC clock source	LSE, LSI or HSE divided by 2 to 511	LSE, LSI, or HSE/ 32
Kernel clock	Independent kernel clock for each peripheral, allowing frequency scaling without impact on communication interfaces	Independent kernel clock for each peripheral, allowing frequency scaling without impact on communication interfaces
System clock source	HSIDIV3, HSI, PSIS or HSE	HSI, CSI, HSE, or PLL1
Clock security system	CSS on HSE, CSS on LSE	CSS on HSE, CSS on LSE
MCO clock source	- MCO1 pin (PA8): SYSCLK, HSE, LSE, LSI, PSIK, HSIK, PSIS or HSI - MCO2 pin (PC9): SYSCLK, HSE, LSE, LSI, PSIK, HSIK, PSIDIV3 or HSIDIV3	- MCO1 pin (PA8): HSI, LSE, HSE, PLL1, or HSI48 - MCO2 pin (PC9): SYSCLK, PLL2, HSE, PLL1, CSI, or LSI

Peripheral clock configuration

The peripherals presented below have a dedicated clock source that is used to generate the clock required for their operations. This section presents the differences between STM32C5 and STM32H523/533 devices, for peripherals with different clock sources.

Table 16. STM32C5 kernel clock distribution overview

Peripherals	Clock multiplexer control bits	Bus clocks	psis_ck	psdiv3_ck	psik_ck	hsis_ck	hsdiv3_ck	hsik_ck	hse_ck	lse_ck	lsi_ck	AUDIOCLK	hse_1M_ck	Disabled
XSPI1	XSPI1SEL	0	-	-	1	-	-	2	-	-	-	-	-	-
FDCANx	FDCANSEL	0	1	-	2	-	-	-	3	-	-	-	-	-
I2Cx	I2CxSEL	0	-	-	1	-	-	2	-	-	-	-	-	-
I3C1	I3C1SEL	0	-	-	1	-	-	2	-	-	-	-	-	-
LPTIM1	LPTIM1SEL	0	-	-	-	-	-	1	-	2	3	-	-	-

Peripherals	Clock multiplexer control bits	Bus clocks	psis_ck	psdiv3_ck	psik_ck	hsis_ck	hsdiv3_ck	hsik_ck	hse_ck	lse_ck	lsi_ck	AUDIOCLK	hse_1M_ck	Disabled
SPI(I2S)x	SPIxSEL	0	-	-	1	-	-	2	-	-	-	3	-	-
USARTx	USARTxSEL	0	-	-	1	-	-	2	-	3	-	-	-	-
UARTx	UARTxSEL	0	-	-	1	-	-	2	-	3	-	-	-	-
LPUART1	LPUART1SEL	0	-	-	-	-	-	1	-	2	3	-	-	-
USB(CK48)	CK48SEL	-	-	1	-	-	2	-	3	-	-	-	-	0
ADC/DAC	ADC/DACSEL	0	1	-	2	-	-	3	-	-	-	-	-	-
DAC	DACSEL	-	-	-	-	-	-	-	-	0	1	-	-	-
RTC/AWU	RTCSEL	-	-	-	-	-	-	-	-	1	2	-	3	0
ETH_CLK	ETHCLKSEL	-	1	-	2	-	-	-	3	-	-	-	-	0
ETH_PTP	ETHPTPCLKSEL	1	2	-	3	-	-	-	-	-	-	-	-	0

Table 17. STM32H5 kernel clock distribution overview

Peripherals ⁽¹⁾	Clock multiplexer control bits	pll1_q_ck	pll2_p_ck	pll2_q_ck	pll2_r_ck	pll3_p_ck	pll3_q_ck	pll3_r_ck	sys_ck	Bus clocks ⁽²⁾	hse_ck	hsi_ker_ck	csi_ker_ck	hsi48_ck	lse_ck	lsi_ck	per_ck ⁽³⁾	AUDIOCLK	Disabled
OCTOSPI	OSPISEL	1	-	-	2	-	-	-	-	0	-	-	-	-	-	-	3	-	-
SDMMC1	SDMMC1SEL	0	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SDMMC2 ⁽⁴⁾	SDMMC2SEL	0	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
FDCAN	FDCANSEL	1	-	2	-	-	-	-	-	-	0	-	-	-	-	-	-	-	-
CEC	CECSEL	-	-	-	-	-	-	-	-	-	-	-	2 ⁽⁵⁾	-	0	1	-	-	-
I2C1	I2C1SEL	-	-	-	-	-	-	1	-	0	-	2	3	-	-	-	-	-	-
I2C2	I2C2SEL	-	-	-	-	-	-	1	-	0	-	2	3	-	-	-	-	-	-
I2C3	I2C3SEL	-	-	-	-	-	-	1	-	0	-	2	3	-	-	-	-	-	-
I2C4	I2C4SEL	-	-	-	-	-	-	1	-	0	-	2	3	-	-	-	-	-	-
I3C1	I3C1SEL	-	-	-	-	-	-	1	-	0	-	2	-	-	-	-	-	-	-
I3C2	I3C2SEL	-	-	-	-	-	-	1	-	0	-	2	-	-	-	-	-	-	-
LPTIM1	LPTIM1SEL	-	1	-	-	-	-	2	-	0	-	-	-	-	3	4	5	-	-
LPTIM2	LPTIM2SEL	-	1	-	-	-	-	2	-	0	-	-	-	-	3	4	5	-	-
LPTIM3	LPTIM3SEL	-	1	-	-	-	-	2	-	0	-	-	-	-	3	4	5	-	-
LPTIM4	LPTIM4SEL	-	1	-	-	-	-	2	-	0	-	-	-	-	3	4	5	-	-
LPTIM5	LPTIM5SEL	-	1	-	-	-	-	2	-	0	-	-	-	-	3	4	5	-	-
LPTIM6	LPTIM6SEL	-	1	-	-	-	-	2	-	0	-	-	-	-	3	4	5	-	-
TIM[8:1],	-	-	-	-	-	-	-	-	-	x	-	-	-	-	-	-	-	-	-
TIM[17:12]	-	-	-	-	-	-	-	-	-	x	-	-	-	-	-	-	-	-	-
TIM16/17	TIMICSEL	-	-	-	-	-	-	-	-	-	-	1	1	-	-	-	-	-	0
RNG	RNGSEL	1	-	-	-	-	-	-	-	-	-	-	-	0	2	3	-	-	-
SAI1	SAI1SEL	0	1	-	-	2	-	-	-	-	-	-	-	-	-	-	4	3	-
SAI2	SAI2SEL	0	1	-	-	2	-	-	-	-	-	-	-	-	-	-	4	3	-

Peripherals ⁽¹⁾	Clock multiplexer control bits	pll1_q_ck	pll2_p_ck	pll2_q_ck	pll2_r_ck	pll3_p_ck	pll3_q_ck	pll3_r_ck	sys_ck	Bus clocks ⁽²⁾	hse_ck	hsi_ker_ck	csi_ker_ck	hsi48_ck	lse_ck	lsi_ck	per_ck ⁽³⁾	AUDIOCLK	Disabled
SPI(I2S)1	SPI1SEL	0	1	-	-	2	-	-	-	-	-	-	-	-	-	-	4	3	-
SPI(I2S)2	SPI2SEL	0	1	-	-	2	-	-	-	-	-	-	-	-	-	-	4	3	-
SPI(I2S)3	SPI3SEL	0	1	-	-	2	-	-	-	-	-	-	-	-	-	-	4	3	-
SPI4	SPI4SEL	-	-	1	-	2	-	-	0	5	3	4	-	-	-	-	-	-	-
SPI5	SPI5SEL	-	-	1	-	2	-	-	0	5	3	4	-	-	-	-	-	-	-
SPI6	SPI6SEL	-	-	1	-	2	-	-	0	5	3	4	-	-	-	-	-	-	-
USARTx	USARTxSEL	-	-	1	-	2	-	-	0	-	3	4	-	5	-	-	-	-	-
UARTx	UARTxSEL	-	-	1	-	2	-	-	0	-	3	4	-	5	-	-	-	-	-
LPUART1	LPUART1SEL	-	-	1	-	2	-	-	0	-	3	4	-	5	-	-	-	-	-
USB	USBSEL	1	-	-	-	2	-	-	-	-	-	-	3	-	-	-	-	-	0
ADCDAC ⁽⁶⁾	ADCDACSEL	-	-	-	2	-	-	-	1	0	3	4	5	-	-	-	-	-	-
DAC	DACSEL	-	-	-	-	-	-	-	-	-	-	-	-	0	1	-	-	-	-
UCPD1	-	-	-	-	-	-	-	-	-	-	-	x ⁽⁷⁾	-	-	-	-	-	-	-
ETH (ptp)	-	x	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
RTC/AWU	RTCSEL	-	-	-	-	-	-	-	-	3 ⁽⁸⁾	-	-	-	1	2	-	-	-	0
All	CKPERSEL	-	-	-	-	-	-	-	-	2	0	1	-	-	-	-	-	-	3

- For the actual availability of the peripheral refer to the device datasheet.
- The bus (APB or AHB) clocks are the bus interface clocks to whom the peripherals are connected.
- The per_ck clock can be hse_ck, hsi_ker_ck, or csi_ker_ck, according to CKPERSEL selection.
- With a duty cycle close to 50%, meaning that PLL1Q or PLL2R values must be even. For SDMMCx the duty cycle must be 50% when supporting DDR
- Clock CSI divided by 122.
- With a duty cycle close to 50%, meaning that PLL2R values must be even.
- Clock HSI divided by 4.
- Clock HSE divided by RTCPRE.

6.2.5 Power (PWR)

The table below presents the PWR controller differences between STM32C5 and STM32H5 devices.

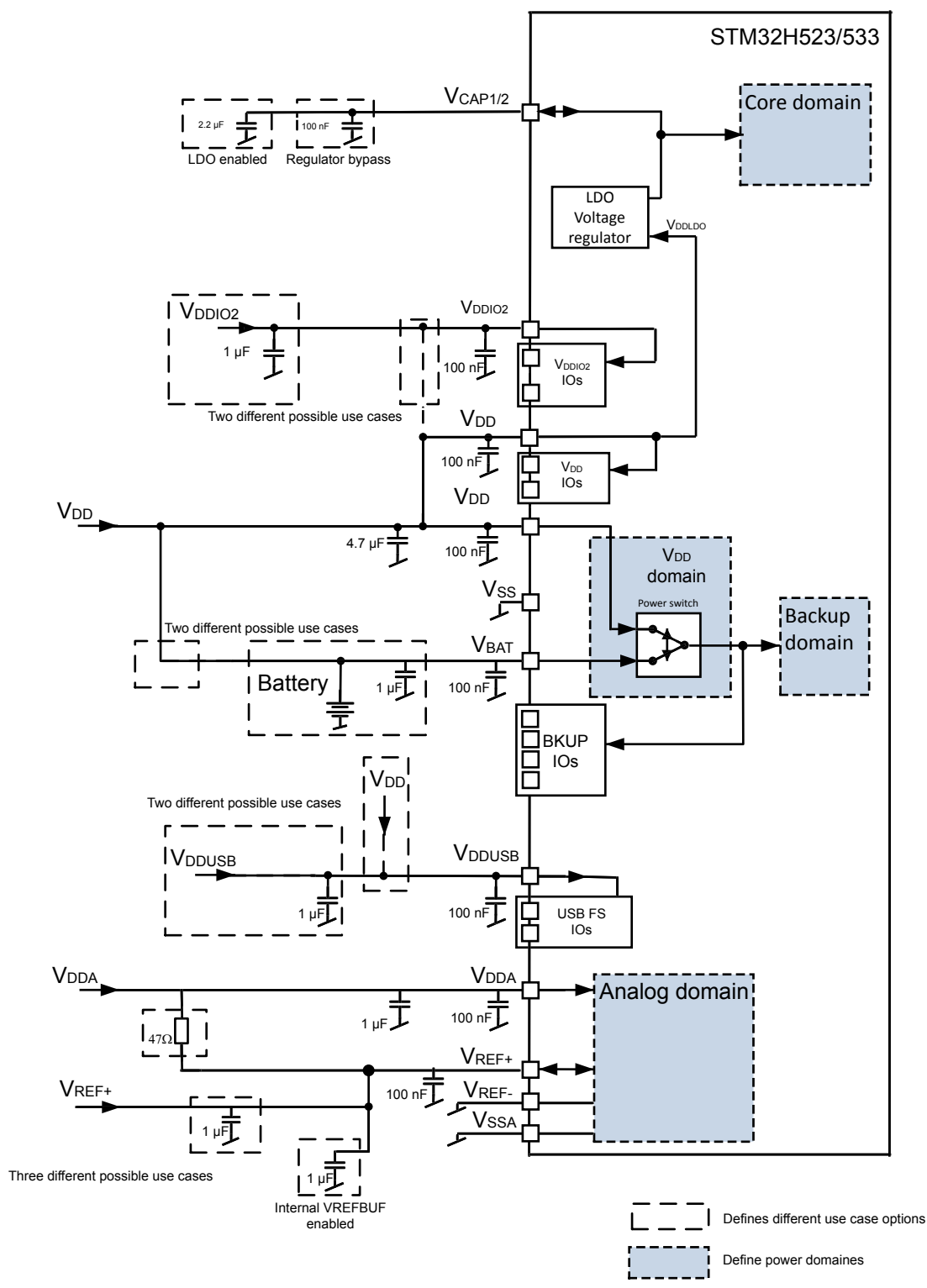
Table 18. PWR features

PWR	STM32H5	STM32C5
Power supplies	V_{DD} = 1.71 V to 3.6 V : external power supply for the I/Os, the internal regulator, and the system analogs such as reset, power management, and internal clocks. It is provided externally through the VDD pins.	V_{DD} = 2.7 to 3.6 V : external power supply for I/Os and the internal regulator (when enabled), provided externally through VDD pins.
	V_{DDA} = 1.62 V ADCs/1.8 V (DAC), 2.1 V (VREFBUF) to 3.6 V : external analog power supply for ADC, DAC, and voltage reference buffer. The VDDA voltage level is independent from the VDD voltage.	V_{DDA} : external analog power supplies for ADC, DAC, reset blocks, RCs. This product has no separated VDDA pad. VDDA is internally connected to VDD.
	V_{CAP} = 1.0 V to 1.35 V : power supply for digital peripherals, SRAMs (except BKPSRAM), and embedded flash memory	V_{CAP} = 0.95 V to 1.25 V : power supply for digital peripherals, SRAMs and embedded flash memory

PWR	STM32H5	STM32C5
Power supplies	V_{BAT} = 1.2 V to 3.6 V : when VDD is not present, VBAT is the power supply for RTC, the external clock 32 kHz oscillator, backup registers, and optional backup SRAM	NA
	V_{DDUSB} = 3.0 V to 3.6 V : external independent power supply for USB transceivers. The VDDUSB voltage level is independent from the VDD voltage.	Supply for USB transceivers. This product has no separated VDDUSB pad. V _{DDUSB} is internally connected to V _{DD}
	V_{DDIO2} = 1.08 V to 3.6 V : external power supply for 10 I/Os (PD6, PD7, PG9:14, PB8, PB9). This voltage is independent from the VDD voltage.	NA
Battery backup domain or RTC domain	RTC with backup registers (128 bytes) LSE LSE, backup SRAM when the low-power backup regulator is enabled PC13 to PC15 I/Os	RTC with backup registers LSE PC13 to PC15 I/Os
Power supply supervisor	POR, PDR, PVD	
	AVD, BOR Backup domain voltage and temperature monitoring	NA
Sleep mode wake-up sources	Any peripheral interrupt/wakeup event	
Standby mode, wake-up sources	WKUPx pin edge, RTC event, external reset in NRST pin, IWDG reset, BOR reset	WKUP pin edge RTC alarm (alarm A and alarm B), RTC wake-up, tamper event, time stamp event, external reset in NRST pin IWDG reset External reset in NRST pin
Stop mode, wake-up sources	Any EXTI line (configured in the EXTI registers) Specific peripheral events	Any EXTI line (configured in the EXTI registers, internal and external lines)
Wake-up system clock	Stop: CSI when STOPWUCK = 1 in RCC_CFGR HSI with the frequency before entering the Stop mode, up to 64 MHz, when STOPWUCK Standby: HSI clock at 64 MHz	Stop: HSI oscillator
Low-power modes	Sleep mode	Sleep mode
	Stop mode: To further optimize the power consumption, the unused RAMs can be totally or partially shut off.	Stop mode to further optimize the power consumption, the unused RAMs can be totally or partially shut off
	Standby mode	Standby mode

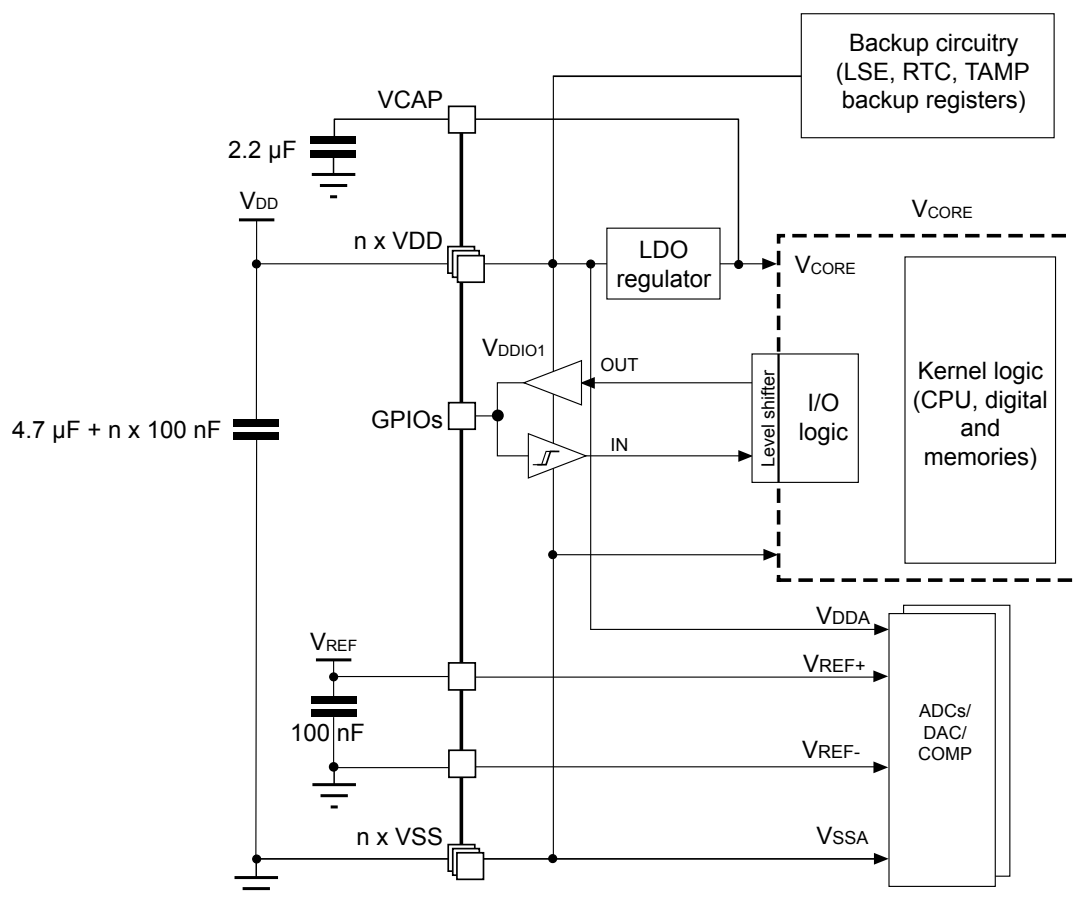
STM32H523/533 devices embed an LDO regulator to provide the V_{CORE} supply for digital peripherals, SRAM1, SRAM2, SRAM3, and the embedded flash memory. This regulator can provide four different voltages (voltage scaling and can operate in stop modes. There is no voltage scaling in C5, LDO regulator can operate under stop mode.

The following figures present the power supply examples for 2 products in STM32C5 and STM32H523/533 series.

Figure 3. STM32H523/533 power supply overview


DT73448V2

Figure 4. STM32C562xx power supply overview



6.2.6 General-purpose I/Os (GPIO)

STM32C5 implement the same GPIO features as STM32H5, but with a few differences.

The support of HSLV is removed as minimum supported power supply is 2.7 V in STM32C5, above threshold of HSLV activation. The support of secure configuration is removed as no Trustzone embedded in STM32C5. All GPIO registers can be read and written by privileged and unprivileged access, whatever the security state (secure or nonsecure).

All GPIO registers can be read and written by privileged and unprivileged access.

6.2.7 Extended interrupt and event controller (EXTI)

6.2.7.1 *EXTI main features*

EXTI features are same between STM32H5 and STM32C5 products on the exception of the trustzone support.

The extended interrupts and event controller (EXTI) manages the individual CPU and system wake-up through configurable event inputs. It provides wake-up requests to the power control and generates an interrupt request to the CPU NVIC and events to the CPU event input. For the CPU, an additional event generation block (EVG) is needed to generate the CPU event signal.

EXTI security protection

When security is enabled for an input event, the associated input event configuration and control bits can only be modified and read by a secure access. A nonsecure write access is discarded and a read returns 0. This is not supported by STM32C5 products.

EXTI privilege protection

When privilege is enabled for an input event, the associated input event configuration and control bits can only be modified and read by a privileged access. An unprivileged write access is discarded and a read returns 0.

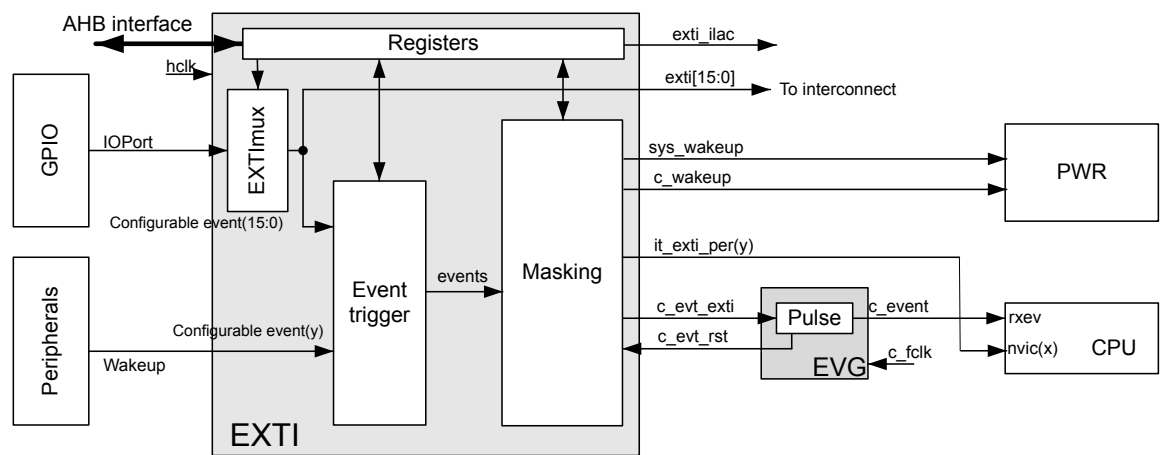
6.2.7.2 EXTI block diagram

As shown in the figure below, EXTI consists of:

- A register block, accessed via an AHB interface
- An event input trigger block
- A masking block, and EXTI mux as shown in the figure below

The register block contains all the EXTI registers. The event input trigger block provides event input edge trigger logic.

Figure 5. EXTI block diagram



The table below presents the EXTI line differences between STM32C5 and STM32H523/533 devices.

Table 19. EXTI line differences

EXTI line	STM32H5	STM32C5
0-15	GPIO	16 external interrupt lines
16	PVD/AVD output	PVD output
17	RTC nonsecure	RTC
18	RTC secure	TAMP
19	TAMP nonsecure	LSE CSS
20	TAMP secure	I2C1 wake-up
21	I2C1 wake-up	I3C1 wake-up
22	I2C2 wake-up	SPI1 wake-up
23	I2C3 wake-up	SPI2 wake-up
24	I3C wake-up	SPI3 wake-up
25	USART1 wake-up	USART1 wake-up
26	USART2 wake-up	USART2 wake-up
27	USART3 wake-up	USART3 wake-up
28	UART4 wake-up	UART4 wake-up

EXTI line	STM32H5	STM32C5
29	UART5 wake-up	UART5 wake-up
30	USART6 wake-up	LPUART1 wake-up
31	Reserved	LPTIM1
32	Reserved	USB FS wake-up
33	Reserved	I2C2 wake-up
34	Reserved	COMP1 OUT
35	Reserved	IWDG early interrupt
36	Reserved	COMP2 OUT
37	LPUART1 wake-up	USART6 wake-up
38	LPTIM1	UART7 wake-up
39	LPTIM2	ETHERNET wake-up
40	SPI1 wake-up	NA
41	SPI2 wake-up	
42	SPI3 wake-up	
43	SPI4 wake-up	
44	Reserved	
45	Reserved	
46	Reserved	
47	USB FS wake-up	
48	USBPD1 wake-up	
49	LPTIM2 CH1	
50	DTS wake-up	
51	HDMI-CEC wake-up	
52	Reserved	
53	UVM output	
54	Reserved	
55	Reserved	
56	Reserved	
57	Reserved	

6.3 Migration of security peripherals

6.3.1 Random number generator (RNG)

STM32C5 and STM32H5 devices embed an RNG that delivers 32-bit random numbers generated by an integrated analog circuit. STM32C5 and STM32H5, the RNG is transparently used by SAES and PKA.

When the RNG finds an unexpected error, an internal tamper event is triggered in the TAMP peripheral, and the RNG stops delivering random data. When this event occurs, the secure application needs to reset the RNG peripheral, using either central reset management or the global SoC reset. After this, another proper initialization of the RNG is required.

6.3.2 Hash processor (HASH)

STM32C5 and STM32H5 devices embed a hash processor (HASH).

HASH is a fully compliant implementation of the secure hash algorithm (SHA-1/SHA-2 family) and the HMAC (keyed-hash message authentication code) algorithm. HMAC is suitable for applications requiring message authentication.

6.3.3 On-the-fly decryption engine (OTFDEC)

The OTFDEC decrypts in real-time the encrypted content stored in the external OCTOSPI memories used in Memory-mapped mode. The OTFDEC uses the AES-128 algorithm in counter mode (CTR).
Only STM32H5 devices embed one OTFDEC peripheral.

Note: This peripheral is not supported by STM32C5.

6.3.4 Public key accelerator (PKA)

STM32C5Ax and STM32H523/H533 devices embed one PKA peripheral, intended for the computation of cryptographic public key primitives within the Montgomery domain.

All needed computations are performed within the accelerator, so no further hardware/software elaboration is needed to process the inputs or the outputs.

6.3.5 AES and SAES hardware accelerators

STM32C5Ax and STM32H523/H533 devices embed two AES accelerators: one secure AES (SAES) and a faster AES.

SAES with a hardware-unique key embeds a protection against differential power analysis (DPA) and related side channel attacks.

When an unexpected hardware fault occurs, an output tamper event is triggered, and AES automatically clears the key registers. A reset is required for AES to function again.

The AES peripheral can use the SAES peripheral as a security coprocessor. In this case, the secure application performs two actions:

- It prepares the key in the robust SAES peripheral.
- When the key is ready, AES can load it through a dedicated hardware key bus.

6.3.6 Global TrustZone® controller (GTZC)

The security architecture of STM32H5 is based on Arm® TrustZone® with the Armv8-M mainline extension.

Each GPIO, peripheral, DMA channel, clock configuration register, DCACHE/ICACHE, or small part of the flash memory or SRAM can be configured as trusted or untrusted.

The GTZC embedded in STM32H5 is used to configure secure TrustZone® and privileged attributes within the full system. A detailed description of GTZC is available in the product reference manual.

Note: GTZC is not supported by STM32C5.

6.4 Migration of communication peripherals

6.4.1 Serial peripheral interface (SPI)

This section highlights the SPI features⁽¹⁾ implemented in STM32C5 and STM32H523/533 devices.

Table 20. SPI features

SPI	STM32H5	STM32C5
Instances	Up to 6x SPIs	3x SPIs
Features	SPI + I2S	
	Including 3 muxed with full-duplex I2S	
Full-duplex synchronous transfer on three lines	X	
Half-duplex	X	
Simplex synchronous transfer on two lines	With a unidirectional data line	
Data size	From 4-bit up to 32-bit data size selection or fixed to multiply of 8-bit	
Multimaster mode capability	X	

SPI	STM32H5	STM32C5
Baud rate prescalers	Baud rate prescaler up to kernel frequency/2 or bypass from RCC in controller mode	
Protection of configuration and settings	X	
Target select (SS) management	Hardware or software management of SS for both controller and target	
Configurable SS signal polarity and timing	Configurable SS signal polarity and timing, MISO x MOSI swap capability	
Programmable transaction data	Programmable number of data within a transaction to control SS and CRC	
Programmable data order with MSB-first or LSB-first shifting	X	
Programmable clock polarity and phase	X	
Dedicated transmission and reception flags with interrupt capability	X	
SPI Motorola and TI format support	X	
Hardware CRC features for reliable communication: - CRC value can be transmitted as the last byte in Tx mode - Automatic CRC error checking for last-received byte	X	
Interrupt events and error detection with interrupt capability	Interrupts: - TxFIFO ready to be loaded - Data received in RxFIFO - Both TXP and RXP are active - Transmission transfer filled - Overrun error - Underrun error - TI frame format error - CRC error - Mode fault - End of transfer - Controller mode suspended - TxFIFO transmission complete All the interrupt events are capable of waking up the system from Sleep mode at each instance	
Configurable behavior at slave-underrun condition	X (support of cascaded circular buffers)	
FIFOs	- Two multiples of 8-bit embedded Rx and Tx FIFOs (FIFO size depends on instance) - Configurable FIFO thresholds (data packing)	
RDY status pin	Optional status pin RDY signals that the target device is ready to handle the data flow	
DRDS delayed read sampling	X	

1. X = available, NA = not available.

6.4.2 Inter-integrated circuit (I2C)

STM32H523/533 devices implement the same I2C features as STM32C5 devices, but with some enhancements. The main differences are stated in the table below.

Table 21. I2C differences

I2C	STM32H5	STM32C5
Instances	x4 (I2C1, I2C2, I2C3 and I2C4)	x3 (I2C1, I2C2, and I2C3)
Features	• 7-bit and 10-bit addressing mode • SMBus/PMBus • Standard mode (up to 100 Kbit/s) • Fast mode (up to 400 Kbit/s)	
	• Fm+ (up to 1 MHz) I2C bus • Wake-up from Stop mode only (no autonomous mode) • Independent clock	

6.4.3 Improved inter-integrated circuit (I3C)

The STM32C5 and STM32H523/533 implement a similar feature set regarding this I3C peripheral.

6.4.4 Universal synchronous/asynchronous receiver transmitter (USART)

STM32H523/533 devices implement several new features of U(S)ART compared to STM32C5 devices. The following table shows the U(S)ART differences.

Table 22. U(S)ART features

USART	STM32H5	STM32C5
Instances	6 USARTs 6 UARTs 1 LPUART	4 USARTs 2 UARTs 1 LPUART
Baud rate	Depends on the frequency (oversampling by 16 or by 8) ⁽¹⁾	
Clock	Dual clock domain and wake-up from low-power mode	
Data	- Word length: programmable (7, 8 or 9 bits) - Programmable data order with MSB-first or LSB-first shifting	
Interrupt	23 interrupt sources with flags	
Others features	- RS232 hardware flow control - RS485 hardware control mode	
	- LIN mode - IrDA SIR encoder block - Continuous communication using DMA - Multiprocessor communications - Single-wire half-duplex communication	
	- Modbus communication: Timeout feature, CR/LF character recognition - Two internal FIFOs for transmit and receive data - Receiver timeout interrupt (except LPUART) - Auto baud rate detection (except LPUART) - Driver enable - Swappable Tx/Rx pin configuration - Wake-up from Stop mode	
	- Smartcard mode: Support the T=0 and T=1 asynchronous protocols - Number of stop bits: 0.5, 1, 1.5, 2	

1. Refer to the USART section in the reference manual.

6.4.5 Controller area network (CAN)

The main differences related to CAN between STM32C5 and STM32H523/533 are presented in the table below.

Table 23. CAN features

CAN	STM32H5	STM32C5
Instances	x2 FDCAN	x2 FDCAN
Features	- Conform with CAN protocol version 2.0-part A, B, and ISO 11898-1: 2015, -4 - CAN FD with maximum 64 data bytes supported - CAN error logging - AUTOSAR and J1939 support - Improved acceptance filtering - Two receive FIFOs of three payloads each (up to 64 bytes per payload) - Separate signaling on reception of high priority messages - Configurable transmit FIFO/queue of three payloads (up to 64 bytes per payload) - Transmit event FIFO - Programmable loop-back test mode - Maskable module interrupts - Two clock domains: APB bus interface and CAN core kernel clock - Power down support - Support time stamping at EOF rather than SOF with a connection to an IC of a timer	

6.4.6 Universal serial bus interface (USB)

The STM32C5 and STM32H523/533 devices have same USB peripherals:

Table 24. USB differences

USB	STM32H5	STM32C5
General	USB FS with clock recovery	
	Up to 8 bidirectional endpoints	
	USB connect/disconnect capability (controllable, embedded pull-up resistor on USB_DP line)	
	Battery charging detection (BCD) support for device	
	Independent VDDUSB power supply	
Buffer memory	2048 bytes of dedicated packet buffer memory SRAM	
Low-power modes	USB revision 2.0 including link power management (LPM) support	

6.5 Migration of analog peripherals

6.5.1 Analog-to-digital converter (ADC)

STM32H523/533 devices embed two ADCs: ADC1 and ADC2. These both consist of a 12-bit successive approximation ADC, are tightly coupled, and can operate in dual mode (ADC1 is the controller).

STM32C5 devices embed three ADCs: ADC1, ADC2, and ADC3 (12-bit resolution).

The table below presents the ADC differences between STM32C5 and STM32H523/533 devices.

Table 25. ADC differences between devices

ADC	STM32H523/533	STM32C5
Instances	x2	x3 on 2 interfaces (ADC12 and ADC3)
Resolution	12-bit	
Number of channels	Up to 20	Up to 28
Configurable resolution	12-bit, 10-bit, 8-bit, or, 6-bit	
Maximum sampling speed	5 MSPS	2.25 MSPS 4.5 MSPS in dual interleaved mode
Conversion modes	• Single • Continuous • Scan • Discontinuous • Dual mode	
DMA support	Yes	
Data register	16-bit data register	
Analog watchdog feature	This feature allows the application to detect if the input voltage goes outside the user-defined high or low threshold	
ADC input range:	$VSSA \leq VIN \leq VREF+$	$VREF- \leq VIN \leq VREF+$
New features	NA	The oversampling is adjustable from 2 to 1024

6.5.2 Digital-to-analog converter (DAC)

STM32H523/533 devices implement some enhanced DAC features compared to STM32C5 devices. Refer to the table below for the main DAC differences between these devices.

Table 26. DAC differences

DAC	STM32H5	STM32C5
Instances	x1 with two output channels	
Resolution	12 bits	
Output buffer	Yes	
Dual DAC channel	For independent or simultaneous conversions	
New features	• Double-data DMA • Buffer offset calibration • Sample and hold mode for low-power operation in Stop mode	

6.6 Migration of timer peripherals

This section compares the features of the above-listed timers and RTC in STM32H523/533 and STM32C5 devices.

Table 27. Merged timer features C5 vs H5

Timer type	Timer in C5	Timer in H5	Counter resolution	Counter type	DMA request generation	Capture / compare channels	Complementary outputs
Advanced control	TIM1, TIM8		16 bits	Up, down, Up/down	Yes	4(+3 internal) in C5 4(+2 internal) in H5	4
General purpose	TIM2, TIM5		32 bits	Up, down, Up/down	Yes	4	No
General purpose	TIM3, TIM4		32 bits in C5 16 bits in H5	Up, down, Up/down	Yes	4	No
General purpose (mid-range)	TIM12		16 bits	Up	No	2	No
	-	TIM13, TIM14	16 bits	Up	No	1	No
General purpose (with complementary)	TIM15		16 bits	Up, down, Up/down in C5 Up in H5	Yes	2	1
General purpose (single-channel with complementary)	TIM16, TIM17		16 bits	Up, down, Up/down in C5 Up in H5	Yes	1	1
Basic	TIM6, TIM7		16 bits	Up	Yes	0	No

6.6.1 Advanced-control timers (TIM1/TIM8)

STM32C5 devices include similar advanced-control timers as STM32H5 series with a few extra features listed below:

Table 28. Advanced-control timer (TIM1/8) features

Feature	STM32H5	STM32C5
Counter resolution and type	16-bit up, down, up/down autoreload counter	

Feature	STM32H5	STM32C5
Prescaler factor	16-bit programmable prescaler, allowing the division (also "on the fly") of the counter clock frequency, by any factor between 1 and 65536	
Channels	Up to six independent channels for: - Input capture (channels 5 and 6) - Output compare - PWM generation (edge- and center-aligned modes) combined PWM mode: 1 and 2 - One-pulse mode output	Up to seven independent channels for: - Input capture (channels 5, 6 and 7) - Output compare - PWM generation (edge- and center-aligned modes) combined PWM mode: 1 to 4 - One-pulse mode output
Complementary outputs	Complementary outputs with programmable dead-time	
Synchronization with external signals and general-purpose timers	- Synchronization circuit to control the timer with external signals and to interconnect several timers together - The advanced-control (TIM1/TIM8) and general-purpose (TIMy) timers are completely independent, and do not share any resources	
Repetition counter	Repetition counter to update the timer registers only after a given number of cycles of the counter	
Break inputs	Two break inputs to put the timer's output signals in reset state or in a known state	Two break inputs to put the timer's output signals in reset state or in a known state featuring delayed break.
Interrupt/DMA generation	Interrupt/DMA generation on the following events: - Update: counter overflow/underflow or counter initialization (by software or internal/external trigger) - Trigger event (counter start, stop, initialization, or count by internal/external trigger) - Input capture - Output compare	
Encoders and sensors	Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes	
Trigger input	Trigger input for external clock or cycle-by-cycle current management	
Application examples	- Measuring the pulse lengths of input signals (input capture) - Generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion)	
Extra features	-	- New extended and asymmetric PWM modes - Output override mode

6.6.2 GP timers with up, down, and up-down autoreload counter (TIM2/3/4/5)

The GP (general-purpose) timers consist of a 16-bit or 32-bit autoreload counter driven by a programmable prescaler.

STM32H5 and STM32C5 devices include GP timers with an up, down, or up/down autoreload counter (TIM2, TIM3, TIM4, and TIM5), with identical features. In STM32C5 TIM3 and TIM4 are 32 bits timers, while they are 16 bits timers in STM32H5 series. Refer to respective datasheet for exact timer list per products.

Table 29. GP timer (TIM2/3/4/5) features

Feature	STM32H5 and STM32C5x
Counter resolution and type	16-bit or 32-bit up, down, up/down autoreload counter
Prescaler factor	16-bit programmable prescaler, used to divide (also “on the fly”) the counter clock frequency by any factor between 1 and 65535
Channels	Up to four independent channels for: • Input capture • Output compare • PWM generation (edge- and center-aligned modes) • One-pulse mode output
Synchronization with external signals and other timers	Synchronization circuit to control the timer with external signals and to interconnect several timers
Interrupt/DMA generation	Interrupt/DMA generation on the following events: • Update: counter overflow/underflow or counter initialization (by software or internal/external trigger) • Trigger event (counter start, stop, initialization, or count by internal/external trigger) • Input capture • Output compare
Encoders and sensors	Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes
Trigger input	Trigger input for external clock or cycle-by-cycle current management
Application examples	• Measuring the pulse lengths of input signals (<i>input capture</i>) • Generating output waveforms (<i>output compare and PWM</i>)

6.6.3 GP timers with autoreload up-counter

STM32H523/533 and STM32C5 devices include 16-bit resolution GP timers with a 16-bit autoreload up-counter: TIM12 and TIM15 are 2 channels timers while TIM13, TIM14, TIM16 and TIM17 are single channel:

- TIM12, TIM13, TIM14, TIM15, TIM16 and TIM17 in STM32H56x/H57x TIM12 and TIM15 in STM32H523/H533 devices
- TIM12, TIM15, TIM16 and TIM17 for STM32C5 devices

Table 30. GP timer (with autoreload up-counter) features

Feature	STM32H523/533	STM32C5
Counter resolution and type	16-bit autoreload up-counter	
Prescaler factor	16-bit programmable prescaler used to divide the counter clock frequency by any factor between 1 and 65535	
Channels	• Input capture • Output compare • PWM generation (edge-aligned mode) • One-pulse mode output	
Complementary outputs	Complementary outputs with programmable dead-time	
Break input	Break input to put the timer's output signals in the reset state or a known state	
Repetition counter	Repetition counter to update the timer registers only after a given number of cycles of the counter	
Interrupt generation	Interrupt/DMA generation on the following events: • Update: counter overflow or counter initialization (by software or internal/external trigger) • Trigger event (counter start, stop, initialization, or count by internal/external trigger) • Input capture • Output compare • Break input (interrupt request)	
Application examples	• Measuring the pulse lengths of input signals (input capture) • Generating output waveforms (output compare, PWM).	

6.6.4 Basic timers (TIM6/7)

The basic timers TIM6 and TIM7 consist of a 16-bit autoreload counter driven by a programmable prescaler. These timers are completely independent, and do not share any resources. STM32H523/533 and STM32C5 devices have the same basic timer features.

Table 31. Basic timers

Feature	STM32C5 and STM32H5
Counter resolution and type	16-bit autoreload up-counter
Prescaler factor	16-bit programmable prescaler, used to divide (also "on the fly") the counter clock frequency by any factor between 1 and 65535
Synchronization signals	Synchronization circuit to trigger the DAC
Interrupt/DMA generation	Interrupt/DMA generation on the update event: counter overflow

6.6.5 Low-power timers (LPTIM1/2)

The LPTIM is a 16-bit timer that benefits from the ultimate developments in power-consumption reduction. Its implementation is similar between STM32C5 with one instance and STM32H5 with two.

Table 32. LPTIM features

Feature	STM32H5	STM32C5
LPTIMx	LPTIM1 and LPTIM2	LPTIM1
Counter resolution and type	16 bit up-counter	
Prescaler factor	3-bit prescaler with 8 possible dividing factors (1,2,4,8,16,32,64,128)	
Selectable clock	- Internal clock sources: configurable internal clock source (see RCC section) - External clock source over LPTIM input (working with no LP oscillator running, used by pulse counter application)	
Autoreload	16 bit ARR auto reload register	
Capture/compare	16 bit capture/compare register	
Continuous mode	Continuous/one-shot mode	
Trigger mode	Selectable software/hardware input trigger	
Glitch filter	Programmable digital glitch filter	
Configurable output	Configurable output: pulse, PWM	
Polarity	Configurable I/O polarity	
Encoder mode	Yes	
Repetition counter	Yes	
Input capture, PWM, and one-pulse channels	Up to two independent channels for: - Input capture - PWM generation (edge-aligned mode) - One-pulse mode output	
DMA requests	DMA request generation on the following events: - Update event - Input capture	

6.6.6 Real-time clock (RTC)

The following table describes the different RTC features between STM32C5 and STM32H523/533 devices. For more information about RTC, refer to the RTC section of the product reference manual.

Table 33. RTC features

RTC	STM32H5	STM32C5
Features	Calendar with subseconds, seconds, minutes, hours (12- or 24-hour format), weekdays, dates, months, and years	
	Two programmable alarms	
	Reference clock detection: a more precise second source clock (50 or 60 Hz) can be used to enhance the calendar precision	
	Timestamp function	
	Daylight saving time	
	Automatic wake-up	
	Digital calibration circuit with 0.95 ppm resolution	
	Alarm A, alarm B, wake-up timer, and timestamp individual privilege protection	
	- Binary mode with 32-bit free-running counter - On-the-fly correction from 1 to 32767 RTC clock pulses - RTC TrustZone® support	- Binary mode with 32-bit free-running counter - On-the-fly correction from 1 to 32767 RTC clock pulses
Tamper and backup registers	32x 32-bit backup registers - Up to 11 tamper pins for 8 external tamper detection events 13 internal tamper events - TrustZone® support	32x 32-bit backup registers - Up to 5 tamper pins for 3 external tamper detection events 6 internal tamper events

6.6.7 SysTick timer

The SysTick timer is dedicated to real-time operating systems, but can also be used as a standard down-counter.

The Cortex®-M33 processor with TrustZone® in STM32H523/533 devices embeds two SysTick timers. When TrustZone® is activated, the two SysTick timers are available, but when TrustZone® is disabled, only one SysTick timer is available.

STM32H5 embeds a Cortex®-M4, with just one SysTick timer.

6.7 Migration of external memory interface peripherals

6.7.1 Octo-SPI interface (OCTOSPI)

The OCTOSPI peripheral implemented in both STM32C59x/Ax and STM32H523/33 provides a serial interface that enables communication with external serial memories such as flash memory, PSRAM, HyperRAM™, and HyperFlash™.

The Octo-SPI specialized communication interface targets single-, dual-, quad-, or octal-SPI memories, and can be configured in three modes: indirect, status-polling, and memory-mapped.

The OCTOSPI peripheral is available on STM32H523/533 and STM32C59x/Ax devices with the following features:

- Functional modes: indirect, status-polling, and memory-mapped
- Read and write support in memory-mapped mode
- Dual-quad configuration
- SDR (single-data rate) and DTR (double-transfer rate)
- Data strobe (DS, DQS)
- GPDMA interface

Revision history

Table 34. Document revision history

Date	Version	Changes
18-Feb-2026	1	Initial release.

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